

Notes (Week 5 Monday)

These are the notes I wrote during the lecture.

Propositional logic has:

boolean variables (propositional atoms)
negation,
conjunction,
disjunction,
implication,
biimplication

^ there's nothing but booleans and combinators
on booleans

In W1, we introduced a logic with three layers:

- terms $+, -, \cup$
- predicates truth claims about terms $\leq, <$
- propositions $\wedge, \vee, \rightarrow$
 $\forall \exists$

$\forall x \in \{1, 2\}. P(x)$
 $\equiv P(1) \wedge P(2)$

^ when the domain is finite,
I can encode universal quantification with
conjunctions

Counterquestion:

what set exactly are x and y from?

Today, quantifiers won't be qualified

$\forall x \in X. P(x)$

we're doing this:

$\forall x (P(x))$

$\forall x y. x = y$

^ holds if the domain of discourse has only one element

otherwise, doesn't hold

There's only ever one domain of discourse.
If you want two, take the union of them.

I want to talk about $\mathbb{N}$ s and $\text{Pow}(\mathbb{R})$ s.
My domain of discourse: $\mathbb{N} \cup \text{Pow}(\mathbb{R})$

$\forall x \in \mathbb{N}. x \geq 0$ $\leftarrow$ I can't write

$\forall x. x \geq 0$ $\leftarrow$ now I'm making a claim also about $\text{Pow}(\mathbb{R})$

$\forall x. \text{is_nat}(x) \rightarrow x \geq 0$
^ added a "is a nat" predicate
to encode quantifier ranges

For exists:

$\exists x \in \mathbb{N}. x < 5$

Encode this as:

$\exists x. \text{is_nat}(x) \wedge x < 5$

Suppose we tried implication instead

$\exists x \in \mathbb{N}. x + 1 = x$

If we used implication instead of conjunction

$\exists x. \text{is_nat}(x) \rightarrow x + 1 = x$

^ this is provable!

Choose $x = \{\pi\}$

$\text{is_nat}(\{\pi\}) \rightarrow \{\pi\} + 1 = \{\pi\}$
 $\equiv \perp \rightarrow \{\pi\} + 1 = \{\pi\}$
 $\equiv \top$

Layers:

terms (constants, functions, variables)
predicates (truth claims on *terms*)
propositions

More appropriate way of saying Jon Snow know nothing:

$\forall x. \text{is_fact}(x) \rightarrow \neg K(J, x)$

In predicate logic, you get to choose some things
about the term structure when you instantiate it.
That's called the vocabulary.

The predicate

`Person("Joe","Biden","The White House")`

^ means that

name: "Joe", surname: "Biden", address: "The White House"
is a row in the Person table

When you specify your vocabulary,
equality always comes "built-in"
Equality is the one predicate that
always has a fixed meaning.

$\forall x P(x) \equiv \forall y P(y)$ if x, y fresh in P
^ doesn't
occur free

$\forall x \exists z \exists x P(x, y, z) \equiv \forall w \exists z \exists x P(x, y, z)$

$\phi = (\forall x \exists z \exists x P(x, y, z)) \wedge Q(x)$

$\phi[1+1/x] = (\forall x \exists z \exists x P(x, y, z)) \wedge Q(1+1)$

$\phi[x/y] \neq (\forall x \exists z \exists x P(x, x, z)) \wedge Q(x) \leftarrow$ that's wrong :(
this illustrates a phenomenon called
variable capture

Solution: rename quantifiers before substituting

$\phi[x/y] = (\forall w \exists z \exists w P(w, x, z)) \wedge Q(x)$

A: It depends upon what the meaning of:
1, 2 and + is.

In propositional logic, this is neither true nor false:

$p \wedge q$

to give it a truth value, we must supply a **valuation**

Vocabulary tells you what symbols you're going to use

The model tells you what they mean

Models give meaning to the **vocabulary**
constants, functions, predicates

But not to variables.

To interpret a formula (to assign truth values to formulas)
we need two things:

interpretation	=	model	+	environment
		^		^
		gives meaning to vocabulary		gives meaning to variables

The reason to treat them separately is:

- the environment may change over the course of a single formula.
 $\forall x. P(x) \wedge \exists y. Q(x,y)$
- when you define e.g. arithmetic (like Giuseppe Peano)
you don't define the values of $x, y, z \dots$ for all time.

η ("eta") stands for an environment

η is a function from variable names
to domain elements

Suppose η is an environment that maps every variable
name to 1.

$$\eta(y) = 1$$

Then $\eta[x \mapsto 2]$ is an environment that maps x to 2,
and otherwise behaves exactly as η

$$\begin{aligned}\eta[x \mapsto 2](x) &= 2 \\ \eta[x \mapsto 2](y) &= 1\end{aligned}$$

$\llbracket \phi \rrbracket^{\eta}_{\mathcal{M}}$ the interpretation of formula ϕ in
the environment η and model $\mathcal{M}$.

$\llbracket t \rrbracket^{\eta}_{\mathcal{M}}$ this is a domain element
^ t for term

$\llbracket \phi \rrbracket^{\eta}_{\mathcal{M}}$ this is a truth value

Let's fix η s.t. $\eta(x) = \text{sandwich}$

Domain: $\{\text{fish}, \text{sandwich}\}$

$\mathcal{M}^1 = \text{fish}$

$\mathcal{M}^+ = f$

such that $f(x, y) = x$

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[[1+x]] = f([[1]], [[x]])  
        = f(fish, sandwich)  
        = fish
```

Isn't it cheating to define $\forall$ in terms of "for all"

I *could* have used a set comprehension instead

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[[ $\forall x \phi$ ]] $\eta, \mathcal{M}$  = true if  
    {c  $\in$  dom( $\mathcal{M}$ ) | [[ $\phi$ ]] $\eta[x \mapsto c], \mathcal{M}$  = true} = dom( $\mathcal{M}$ )
```

But there is a fundamental problem here.

To give meaning to a formal system

(to define the semantics of it)

formal system: propositional logic, predicate logic,
 ZFC set theory,

I need a bigger formal system.

How do I give meaning to *that* system?